AN IDEAL IN AN ABSOLUTELY INTEGRALLY CLOSED RING GIVES A HENSELIAN PAIR UNDER TWO CONDITIONS

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Absolutely integrally closed ring, meaning every monic polynomial decomposes into linear factors. definition environment at source lines 3509-3515 of the same chapter. Supplies the roots a_i of f in A, which are the points at which the idempotents are evaluated.
- Henselian pair, defined by the lifting of factorisations of monic polynomials modulo the ideal. the definition used by the chapter, cited inside the surrounding text of source lines 3580-3618. Fixes the goal: any factorisation modulo I with monic coprime factors must lift.
- Idempotents of a ring correspond to decompositions of the ring as a finite product of rings. used at proof lines 3640-3647 where A is written as a finite product with each e_i equal to 0 or 1. Reduces the problem to the case where each evaluated idempotent is 0 or 1, which is what makes the induction on the degree possible.

Source reference. Original lemma 0DCT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16743–16743. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let A be absolutely integrally closed. Let $I \subset A$ be an ideal. Then (A, I) is a henselian pair if (and only if) the following conditions hold*

(1) *I is contained in the Jacobson radical of A,*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *$A \to A/I$ induces a bijection on idempotents.*

Proof. Let $f \in A[T]$ be a monic polynomial and let $f \bmod I = g_0 h_0$ be a factorization over A/I with g_0, h_0 monic such that g_0 and h_0 generate the unit ideal of $A/I[T]$. This means that

$$A/I[T]/(f) = A/I[T]/(g_0) \times A/I[T]/(h_0)$$

Denote $e \in A/I[T]/(f)$ the element correspoing to the idempotent $(1, 0)$ in the ring on the right. Write $f = (T - a_1) \ldots (T - a_d)$ with $a_i \in A$. For each $i \in \{1, \ldots, d\}$ we obtain an A-algebra map $\varphi_i : A[T]/(f) \to A$, $T \mapsto a_i$ which induces a similar A/I-algebra map $\overline{\varphi}_i : A/I[T]/(f) \to A/I$. Denote $e_i = \overline{\varphi}_i(e) \in A/I$. These are idempotents. By our assumption (2) we can lift e_i to an idempotent in A. This means we can write $A = \prod A_j$ as a finite product of rings such that in A_j/IA_j each e_i is either 0 or 1. Some details omitted. Observe that A_j is absolutely integrally closed as a factor ring of A. It suffices to lift the factorization of f over A_j/IA_j to A_j. This reduces us to the situation discussed in the next paragraph.

Assume $e_i = 1$ for $i = 1, \ldots, r$ and $e_i = 0$ for $i = r + 1, \ldots, d$. From $(g_0, h_0) = A/I[T]$ we have that there are $k_0, l_0 \in A/I[T]$ such that $g_0 k_0 + h_0 l_0 = 1$. We see that $e = h_0 l_0$ and $e_i = h_0(a_i) l_0(a_i)$. We conclude that $h_0(a_i)$ is a unit for $i = 1, \ldots, r$. Since $f(a_i) = 0$ we find $0 = h_0(a_i) g_0(a_i)$ and we conclude that $g_0(a_i) = 0$ for $i = 1, \ldots, r$. Thus $(T - a_1)$ divides g_0 in $A/I[T]$, say $g_0 = (T - a_1) g_0'$. Set $f' = (T - a_2) \ldots (T - a_d)$ and $h_0' = h_0$. By induction on d we can lift the factorization $f' \bmod I = g_0' h_0'$ to a factorization of $f' = g' h'$ over over A which gives the factorization $f = (T - a_1) g' h'$ lifting the factorization $f \bmod I = g_0 h_0$ as desired. $\square$